

Experiment 3: Crypt Arithmetic Problem

Aim

To implement and solve the Crypt Arithmetic Problem using Artificial Intelligence techniques.

Algorithm

1. Start
2. Initialize the problem state
3. Apply valid operations/search
4. Repeat until goal state is reached
5. Display solution
6. Stop

Flowchart

```
graph TD
    Start([Start]) --> Init[Initialize]
    Init --> Process[Process/Search]
    Process --> Goal{Goal Reached?}
    Goal -- Yes --> Print[Print]
    Goal -- No --> Repeat[Repeat]
    Repeat --> Process
    Print --> Stop([Stop])
```

Objective

- Understand state-space search.
- Implement AI search technique.
- Obtain the required goal state.

Program

```
from itertools import permutations
```

```
for s,e,n,d,m,o,r,y in permutations(range(10),8):
```

```
    if s and m:
```

```
        send = 1000*s+100*e+10*n+d
```

```
        more = 1000*m+100*o+10*r+e
```

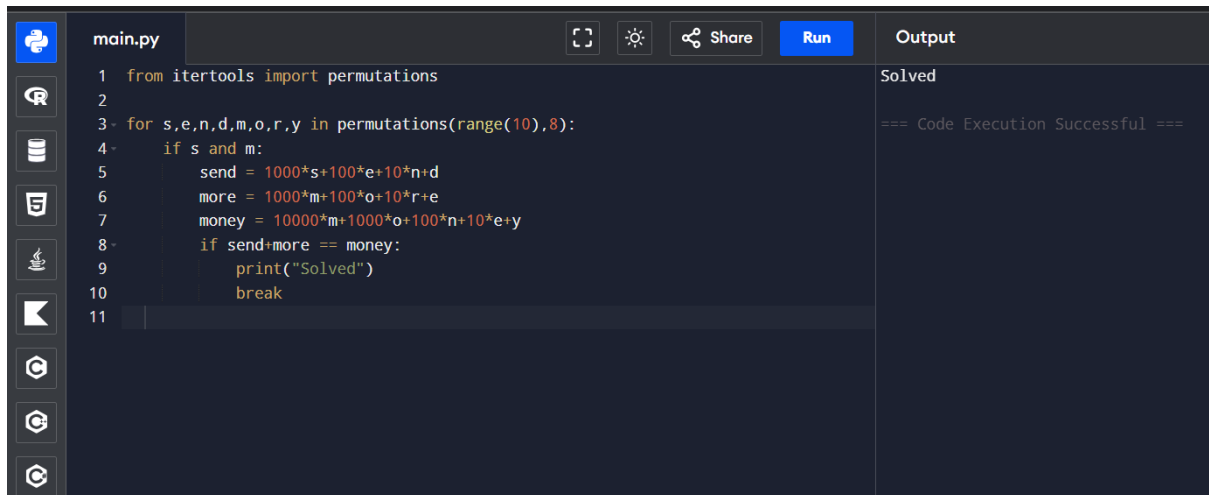
```
money = 10000*m+1000*o+100*n+10*e+y
```

```
if send+more == money:
```

```
    print("Solved")
```

```
    break
```

Output



The screenshot shows a Jupyter Notebook interface with a dark theme. On the left is a sidebar with icons for file explorer, search, and other tools. The main area is split into two panes. The left pane, titled 'main.py', contains the following Python code:

```
1 from itertools import permutations
2
3 for s,e,n,d,m,o,r,y in permutations(range(10),8):
4     if s and m:
5         send = 1000*s+100*e+10*n+d
6         more = 1000*m+100*o+10*r+e
7         money = 10000*m+1000*o+100*n+10*e+y
8         if send+more == money:
9             print("Solved")
10            break
11
```

The right pane, titled 'Output', displays the result of the code execution:

```
Solved
=== Code Execution Successful ===
```

Result

The program was executed successfully.

Conclusion

The program demonstrated the AI concept successfully and produced the expected result.